

STeM: Symbolic Tetrad Manipulator. Part I

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Abstract

In this first part, of a two-part presentation, we present and describe, with illustrative examples, some of the basic functionality of the Maple computer algebra package *STeM*, which has a comprehensive collection of routines to allow the user to carry out most computations in the Newman-Penrose, Ellis-MacCallum, and Geroch-Held-Penrose formalisms, or any other tetrad formalism which is a modification of any one of these three systems.

1 Program summary

Program Title: STeMV1R1

CPC Library link to program files:

Developer's repository link: <https://github.com/jcarm1930/STeMV1R1>

Licensing provisions: CC BY-ND

Programming language: Maple internal language

Nature of Problem: The tetrad formalism is an approach to general relativity that generalizes the choice of basis for the tangent bundle from a coordinate basis to a more general choice of a local basis, i.e. a locally defined set of four linearly independent vector fields called a tetrad.¹¹ Usually the choice of the tetrad is guided by the geometric/physical properties of the problem at hand. The advantage of the tetrad formalism over the standard coordinate-based approach to general relativity lies in the ability to choose the tetrad basis to reflect important physical/geometric aspects of the spacetime. This can be quite useful when investigating solutions and their properties, of the Einstein field equations. In the tetrad formalism, all tensors are represented in terms of their components (scalars) in a chosen basis and computations are conveniently carried out in terms of these components. Since the tensorial equations are written in terms of their basis components, large expressions can arise. This can be an advantage as well as a drawback but in many cases, however, the properties that one seeks or assumes usually become more evident. Apart from the direct use of the Ricci identities, Einstein field equations and Bianchi identities, one usually extracts important additional information through the use of the commutator conditions present in all the three major formalisms, Newman-Penrose(NP),⁹ Ellis-MacCallum(EM)^{3 7} and Geroch-Held-Penrose(GHP),⁵ mainly used in General Relativity. Usually, these combined tetrad computations and resulting scalar equations, can be extensive and complicated, well beyond hand computation ability. In this regard, we have designed *STeM* to include interactive tools to assist the user in the generation and analysis of such systems.

Solution method: *STeM*, which runs under Maple, consists of a comprehensive collection of routines and complete set of equations, to allow the user to carry out most complex calculations and analyses in the three formalisms NP, EM, GHP or any other tetrad formalism which is a modification of any one of these three systems.

Keywords: Computer Algebra, General Relativity, Exact solutions, Tetrad formalisms.

2 Introduction: Tetrad Formalisms

In general relativity's original inception, the field equations (EFE) were written by Einstein in tensor form in a coordinate basis. In time and with the development of more sophisticated mathematics especially with the works of E. Cartan, it was recognised that the analyses, and subsequent clarification, of geometric aspects in the theory were more easily carried out with the introduction of a noncoordinate basis. Great impetus to this new approach was given by the seminal paper by Newman and Penrose (NP)⁹ in 1962, where with the use of spinors and spin bases, the spin coefficient, null tetrad approach was basically born. In their approach, at each point

of the spacetime, a complex null tetrad $\{\mathbf{l}, \mathbf{n}, \mathbf{m}, \bar{\mathbf{m}}\}$, with the full freedom of choice given by the 6-parameter Lorentz group, was introduced with the orthogonality properties

$$\begin{aligned} l_a l^a &= n_a n^a = m_a m^a = l_a m^a = n_a m^a = 0, \\ l_a n^a &= -m_a \bar{m}^a = 1, \end{aligned}$$

and tetrad metric

$$\eta_{ab} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

so that

$$g_{ab} = 2l_{(a}n_{b)} - 2m_{(a}\bar{m}_{b)}.$$

In applying the NP "method", one usually aligns the tetrad according to the symmetries of the spacetime or with the eigenvectors singled out by a particular tensor like the Weyl or Maxwell tensor with the consequence that substantial simplifications occur. The method has proven to be quite useful in a number of classification problems like those of the Weyl and Ricci tensors. We note that important solutions of the EFE were (and still are) quickly obtained, in particular the rotating Kerr solution, after the appearance of the NP formalism. This formalism is currently, more than ever, used in a wide range of applications with considerable success.

Sometime later, it was in particular noted by G. F. R. Ellis, M.A.H. MacCallum and others that the introduction of a null tetrad was not really suited to studying fluid sources and cosmological models of the EFE. As a consequence of their work, it became clear that a formulation of the EFE using orthonormal tetrads instead would be more appropriate. Considerable impetus to this approach was given by the seminal papers of Ellis³ and MacCallum.⁷ In these spacetimes, there is a time-like unit vector $\mathbf{u}$ singled out by the fluid flow. Hence it seemed more appropriate to introduce an orthonormal tetrad $\{\mathbf{t}, \mathbf{x}, \mathbf{y}, \mathbf{z}\}$ at each point of the spacetime, with the orthogonality properties

$$\begin{aligned} t_a x^a &= t_a y^a = t_a z^a = x_a y^a = x_a z^a = y_a z^a = 0, \\ -t_a t^a &= x_a x^a = y_a y^a = z_a z^a = 1, \end{aligned}$$

and tetrad metric

$$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

so that

$$g_{ab} = -t_a t_b + x_a x_b + y_a y_b + z_a z_b.$$

In applying this approach, one usually aligns the vector $\mathbf{t}$ with $\mathbf{u}$. One then projects all physical (geometric) quantities of interest, to spaces parallel and orthogonal to $\mathbf{u}$. In doing so, one obtains spacelike tensors/components like the vorticity, shear, expansion, of the fluid, and electric and magnetic parts of the Weyl tensor and so on. The tetrad is then further aligned (specialised), with the additional freedom of the $SO(3)$ group, with these quantities, their eigenvectors, or with any symmetries present.

In 1973, R. Geroch, A. Held, and R. Penrose,⁵ introduced an interesting variant of the Newman-Penrose formalism, where a pair of null directions was singled out. It is an intermediate formalism between a fully covariant one and the spin-coefficient approach. The key idea is to only retain quantities in the NP formalism which are properly weighted under the spin-boost 2-parameter subgroup of the Lorentz group. Of the original 12 complex spin coefficients only 8 remain with modifications of the derivative operators and commutator conditions. A remarkable, considerable simplification occurs. This formalism has been quite useful in the study/integration of algebraically special spacetimes or with the analysis of perturbations of them.

Regarding the *STeM* package, it is made up of three sub-systems corresponding to the three formalisms mentioned above. The orthonormal implementation follows closely the basic functionality and notation as that given by N. Van den Bergh in his program called *Oframe*.¹³ The NP part follows the functionality and notation as that given by S Czapor and R.G. McLenaghan in their NP package.² Finally, The GHP part which includes all of the routines in our previous package is an expansion of our previous GHP package.¹⁵ For a very comprehensive list of computer algebra packages for general relativity, we refer the reader to the article *Computer algebra in gravity research* by M.A. H. MacCallum.⁸

3 The Maple STeM Package: Notation and Conventions and Comparisons

Throughout, *STeM* uses the $(-, +, +, +)$ signature with Levi-Civita tensor and permutation symbol defined by $\eta^{0123} = (-g)^{-1/2}$, and $\epsilon_{123} = 1$. The Riemann, Ricci, and Weyl tensors are defined by

$$P^a{}_{;cd} - P^a{}_{;dc} = R^a{}_{bdc}P^b, R_{ab} = R^c{}_{acb},$$

$$C_{abcd} = R_{abcd} - \frac{1}{2}(g_{ac}R_{bd} + g_{bd}R_{ac} - g_{bc}R_{ad} - g_{ad}R_{bc}) + \frac{R}{6}(g_{ac}g_{bd} - g_{ad}g_{bc}).$$

where P is an arbitrary vector field.

The electric and magnetic parts of the Weyl tensor are defined by

$$E_{ab} = C_{agbh}u^gu^h, H_{ab} = \frac{1}{2}\eta_{ae}{}^{gh}C_{ghbd}u^eu^d.$$

Given an electromagnetic field represented by a skew tensor field F_{ab} , the electric field E_a and magnetic field H_a , as measured by an observer with 4-velocity u^a are given by

$$E_a = F_{ab}u^b, H_a = \frac{1}{2}\eta_{ab}{}^{cd}u^bF_{cd}.$$

The relations between the NP spin coefficients and the orthonormal quantities are⁶

$$\begin{aligned}\alpha &= \frac{1}{2\sqrt{2}}\{a_2 + n_{13} - \sigma_{12} + \omega_3 + i(n_{12} + \omega_2 - \sigma_{13} - a_3)\}, \beta = \frac{1}{2\sqrt{2}}\{\sigma_{12} + \omega_3 + i(n_{12} + \sigma_{13} - a_3 - \omega_2)\} \\ \epsilon &= \frac{1}{4\sqrt{2}}\{2(\dot{u}_1 + \theta_1) + i(n_{11} - 2\Omega_1 - n_{22} - n_{33})\}, \gamma = \frac{1}{4\sqrt{2}}\{2(\dot{u}_1 - \theta_1) + i(n_{22} - n_{11} + n_{33} + \Omega_1)\}, \\ \kappa &= \frac{1}{2\sqrt{2}}\{n_{13} - \sigma_{12} - \dot{u}_2 + \omega_3 - a_2 - \Omega_3 + i(\Omega_2 - a_3 - n_{12} - \sigma_{13} - \dot{u}_3 - \omega_2)\}, \\ \lambda &= \frac{1}{2\sqrt{2}}\{2n_{23} + \theta_2 - \theta_3 + i(n_{22} - n_{33} - 2\sigma_{23})\}, \mu = \frac{1}{2\sqrt{2}}\{2a_1 + \theta_2 + \theta_3 + i(2\omega_1 - n_{11})\}, \\ \nu &= \frac{1}{2\sqrt{2}}\{a_2 - n_{13} - \sigma_{12} - \Omega_3 + \dot{u}_2 + \omega_3 + i(\sigma_{13} + \omega_2 - \dot{u}_3 - n_{12} + a_3 - \Omega_2)\}, \\ \pi &= \frac{1}{2\sqrt{2}}\{n_{13} + \sigma_{12} + \dot{u}_2 - \omega_3 - \Omega_3 - a_2 + i(n_{12} + a_3 - \Omega_2 - \sigma_{13} - \dot{u}_3 - \omega_2)\}, \\ \varrho &= \frac{1}{2\sqrt{2}}\{2a_1 - \theta_2 - \theta_3 + i(n_{11} + 2\omega_1)\}, \sigma = \frac{1}{2\sqrt{2}}\{2n_{23} + \theta_3 - \theta_2 + i(n_{33} - n_{22} - 2\sigma_{23})\}, \\ \tau &= \frac{1}{2\sqrt{2}}\{a_2 + \sigma_{12} - \Omega_3 - n_{13} - \dot{u}_2 - \omega_3 + i(n_{12} - \dot{u}_3 + a_3 + \Omega_2 + \omega_2 + \sigma_{13})\}.\end{aligned}$$

The NP (Gaussian units) and orthonormal (Heaviside-Lorentz units) forms of Maxwell's equations are

$$\begin{aligned}D\Phi_1 - \bar{\delta}\Phi_0 + \kappa\Phi_2 - 2\rho\Phi_1 + (2\alpha - \pi)\Phi_0 &= x_0J_l, \\ \delta\Phi_1 - \Delta\Phi_0 + \sigma\Phi_2 - 2\tau\Phi_1 + (2\gamma - \mu)\Phi_0 &= x_0J_m, \\ D\Phi_2 - \bar{\delta}\Phi_1 + (2\epsilon - \rho)\Phi_2 - 2\pi\Phi_1 + \lambda\Phi_0 &= x_0J_{\bar{m}}, \\ \delta\Phi_2 - \Delta\Phi_1 + (2\beta - \tau)\Phi_2 - 2\mu\Phi_1 + \nu\Phi_0 &= x_0J_n. \\ \nabla \cdot \mathbf{E} &= (\mathbf{N} - \mathbf{a}) \cdot \mathbf{E} + 2\boldsymbol{\omega} \cdot \mathbf{B} - j_0 \\ \nabla \cdot \mathbf{B} &= (\mathbf{N} - \mathbf{a}) \cdot \mathbf{B} - 2\boldsymbol{\omega} \cdot \mathbf{E}, \\ \mathbf{e}_0(\mathbf{E}) &= \mathbf{e} \times \mathbf{B} - \hat{j} - \frac{2}{3}\theta\mathbf{E} + \mathbf{N}[\mathbf{B}] - \sigma[\mathbf{E}] + \boldsymbol{\omega} \times \mathbf{E} \\ \mathbf{e}_0(\mathbf{B}) &= -\mathbf{e} \times \mathbf{E} - \frac{2}{3}\theta\mathbf{B} - \mathbf{N}[\mathbf{E}] - \sigma[\mathbf{B}] + \boldsymbol{\omega} \times \mathbf{B}\end{aligned}$$

where x_0 stands for the numerical constant 2π , and where

$$\begin{aligned}
J^a &= J_m \bar{m}^a + J_{\bar{m}} m^a - J_l n^a - J_n l^a, (Q \times R)_i = \epsilon_{ijk} Q_j R_k, \\
E_a + iH_a &= \sqrt{2} \Phi_1 (l_a - n_a) + \sqrt{2} (\Phi_2 m_a - \Phi_0 \bar{m}_a), \\
\mathbf{E} + i\mathbf{H} &= 2\Phi_1 \mathbf{e}_1 + (\Phi_2 - \Phi_0) \mathbf{e}_2 + i(\Phi_0 + \Phi_2) \mathbf{e}_3, \\
\mathbf{e}_0 &= \frac{1}{\sqrt{2}} (\mathbf{l} + \mathbf{n}), \mathbf{e}_1 = \frac{1}{\sqrt{2}} (\mathbf{l} - \mathbf{n}), \mathbf{e}_2 = \frac{1}{\sqrt{2}} (\mathbf{m} + \bar{\mathbf{m}}), \mathbf{e}_3 = \frac{-i}{\sqrt{2}} (\mathbf{m} - \bar{\mathbf{m}}), \\
E_1 &= (\Phi_1 + \bar{\Phi}_1), E_2 = \frac{1}{2} (\Phi_2 + \bar{\Phi}_2 - \Phi_0 - \bar{\Phi}_0), E_3 = \frac{i}{2} (\Phi_2 - \bar{\Phi}_2 + \Phi_0 - \bar{\Phi}_0), \\
H_1 &= i(\bar{\Phi}_1 - \Phi_1), H_2 = \frac{i}{2} (\bar{\Phi}_2 - \Phi_2 + \Phi_0 - \bar{\Phi}_0), H_3 = \frac{1}{2} (\Phi_0 + \bar{\Phi}_0 + \Phi_2 + \bar{\Phi}_2), \\
\Phi_0 &\equiv F_{ab} l^a m^b = \phi_{11}, \Phi_1 \equiv \frac{1}{2} F_{ab} (l^a n^b + \bar{m}^a m^b) = \phi_{01}, \Phi_2 \equiv F_{ab} \bar{m}^a n^b = \phi_{00}, \\
J_l &\equiv J_a l^a, J_n \equiv J_a n^a, J_m \equiv J_a m^a, J_{\bar{m}} \equiv J_a \bar{m}^a = \bar{J}_m, (\mathbf{e} \times \mathbf{P})_i \equiv \epsilon_{ijk} \mathbf{e}_j (B_k), \\
N_i &\equiv n_{i1} + n_{i2} + n_{i3}, \mathbf{N} \equiv N_1 \mathbf{e}_1 + N_2 \mathbf{e}_2 + N_3 \mathbf{e}_3, \hat{\mathbf{j}} = j_1 \mathbf{e}_1 + j_2 \mathbf{e}_2 + j_3 \mathbf{e}_3 \\
\mathbf{a} &\equiv a_1 \mathbf{e}_1 + a_2 \mathbf{e}_2 + a_3 \mathbf{e}_3, \sigma[\mathbf{P}]_a \equiv \sigma_{ab} P^b, \mathbf{j} = -j_0 \mathbf{e}_0 + \hat{\mathbf{j}}
\end{aligned}$$

The vector $\sigma[\mathbf{P}]$ is the anisotropic distortion of the vector $\mathbf{P}$ produced by the shear of the time-like congruence and $\mathbf{N}[\mathbf{P}]$ is the distortion induced on $\mathbf{P}$ purely by the non-integrability and anisotropy of the spatial tetrad. With geometrized units Einstein's equations read $G_{ab} = 4x_0 T_{ab}$. The electromagnetic contribution to the NP components of the Ricci tensor is $\Phi_{ab} = \Phi_a \bar{\Phi}_b$, with $\Lambda = 0$. With our signature we have $l_a n^a = -1, m_a \bar{m}^a = 1$, and the Weyl and Ricci NP scalars need to change sign, to those given by NP, in order to maintain the same form for the Einstein- Maxwell equations. These quantities are explicitly given below:

$$\begin{aligned}
\Psi_0 &\equiv C_{abcd} l^a m^b l^c m^d, \Psi_1 \equiv C_{abcd} l^a n^b l^c m^d, \\
\Psi_2 &\equiv C_{abcd} l^a m^b \bar{m}^c n^d, \Psi_3 \equiv C_{abcd} l^a n^b \bar{m}^c n^d, \\
\Psi_4 &\equiv C_{abcd} n^a \bar{m}^b n^c \bar{m}^d,
\end{aligned}$$

and

$$\begin{aligned}
\Phi_{00} &\equiv \frac{1}{2} R_{ab} l^a l^b, \Phi_{11} \equiv \frac{1}{4} R_{ab} (l^a n^b + m^a \bar{m}^b), \Phi_{22} \equiv \frac{1}{2} R_{ab} n^a n^b, \Lambda \equiv \frac{R}{24}, \\
\Phi_{01} &= \bar{\Phi}_{10} \equiv \frac{1}{2} R_{ab} l^a m^b, \Phi_{02} = \bar{\Phi}_{20} \equiv \frac{1}{2} R_{ab} m^a m^b, \\
\Phi_{12} &= \bar{\Phi}_{21} \equiv \frac{1}{2} R_{ab} m^a n^b.
\end{aligned}$$

The three, 2-parameter null subgroups of the Lorentz group acting on the null basis vectors $\{l, n, m, \bar{m}\}$ are given below.

Null rotation leaving l^a fixed:

$$\begin{aligned}
\tilde{l}^a &= l^a, \\
\tilde{n}^a &= n^a + d\bar{m}^a + \bar{d}m^a + d\bar{d}l^a, \\
\tilde{m}^a &= m^a + dl^a.
\end{aligned}$$

Null rotation leaving n^a fixed:

$$\begin{aligned}
\tilde{n}^a &= n^a, \\
\tilde{l}^a &= l^a + f\bar{m}^a + \bar{f}m^a + f\bar{f}n^a, \\
\tilde{m}^a &= m^a + fn^a.
\end{aligned}$$

Null rotation leaving directions of l^a and n^a fixed

$$\begin{aligned}
\tilde{l}^a &= F^{-1} l^a, \\
\tilde{n}^a &= F n^a, \\
\tilde{m}^a &= e^{i\theta} m^a.
\end{aligned}$$

The notation used for each formalism is given in the tables below.

Newman-Penrose Formalism: Notation

Original	STeM
D	D
Δ	V
δ	X
$\bar{\delta}$	Y
Ψ_0	$W0$
Ψ_1	$W1$
Ψ_2	$W2$
Ψ_3	$W3$
Ψ_4	$W4$
Λ	L
Φ_{00}	$R00$
Φ_{01}	$R01$
Φ_{02}	$R02$
Φ_{12}	$R12$

Original	STeM
Φ_{11}	$R11$
Φ_{22}	$R22$
κ	k
σ	s
ρ	r
τ	t
ν	n
λ	l
μ	m
π	p
ϵ	e
γ	g
α	a
β	b

Original	STeM
Φ_0	$F0$
Φ_1	$F1$
Φ_2	$F2$
V^1	Kl
V^2	Kn
V^3	Km
ψ	psi
ϕ_{00}	$B00$
ϕ_{10}	$B10$
ϕ_{11}	$B11$
J_l	$J0$
J_n	$J1$
J_m	$J2$

Ellis-MacCallum Formalism: Notation

Original	STeM	Original	STeM	Original	STeM
e_0	$e0$	n_{12}	$n12$	$\theta_1 = \theta_{11}$	$H1$
e_1	$e1$	n_{13}	$n13$	$\theta_2 = \theta_{22}$	$H2$
e_2	$e2$	n_{22}	$n22$	$\theta_3 = \theta_{33}$	$H3$
e_3	$e3$	n_{23}	$n23$	θ	$H = H1 + H2 + H3$
a_1	$a1$	n_{33}	$n33$	μ	M
a_2	$a2$	σ_{12}	$s12$	p	P
a_3	$a3$	σ_{13}	$s13$	$E_{\alpha\beta}$	$E11..E33$
$\dot{u}_1$	$u1$	σ_{23}	$s23$	$H_{\alpha\beta}$	$M11..M33$
$\dot{u}_2$	$u2$	V^1	$K0$	$E_{ab;c}u^c$	$dE11..$
$\dot{u}_3$	$u3$	V^2	$K1$	$H_{ab;c}u^c$	$dM11..$
ω_1	$w1$	V^3	$K2$	E_α	$E1..E3$
ω_2	$w2$	V^4	$K3$	H_α	$B1..B3$
ω_3	$w3$	Ω_1	$O1$	j_0	$j0$
n_{11}	$n11$	Ω_2	$O2$	j_1	$j1$
n_{12}	$n12$	Ω_3	$O3$	j_2	$j2$
				j_3	$j3$

Geroch-Held-Penrose: Notation

Original	STeM
$\mathfrak{p}$	T
$\mathfrak{p}'$	TP
δ	E
δ'	EP

In our implementation, the NP components of the Killing (Homothetic) vector are given by¹⁰

$$K^a = Km\bar{m}^a + K\bar{m}m^a - Kl n^a - Kn l^a, \quad K\bar{m} \equiv \overline{Km}$$

whereas in the orthonormal frame, the Killing vector $\mathbf{K}$ is expressed as

$$\begin{aligned} \mathbf{K} &= -K0\mathbf{e}_0 + K1\mathbf{e}_1 + K2\mathbf{e}_2 + K3\mathbf{e}_3, \\ Kl &= \frac{1}{\sqrt{2}}(K0 + K1), \quad Kn = \frac{1}{\sqrt{2}}(K0 - K1), \quad K\bar{m} = \frac{1}{\sqrt{2}}(K2 + iK3) \end{aligned}$$

and the Killing equations are written as

$$K_{a;b} = Z_{ab} + \psi g_{ab},$$

where $\psi = 0$ for a Killing vector or $\psi = \text{constant} \neq 0$ for a homothetic vector.

The associated complex self-dual bivector is

$$\begin{aligned} Z_{ab}^+ &\equiv \frac{1}{2}(Z_{ab} + iZ_{ab}^*) \\ &= 2\phi_{00}l_{[a}m_{b]} + 2\phi_{01}(l_{[a}n_{b]} - m_{[a}\bar{m}_{b]}) - 2\phi_{11}n_{[a}\bar{m}_{b]} \end{aligned}$$

A list of current *STeM* package routines, for each formalism with short descriptions is given in the tables below.

Available Commands, with Syntax, for all Formalisms

Command/Syntax	Description
addpair(sys)	adds and factors all distinct pair additions of polys in list sys
allddeg(sys)	displays the degree of each variable in sys
autocomm(vars)	computes all 6 basic commutators of vars
buildchain(sys,chn,R)	constructs a new chain from inclusion of sys
chaincombo(PB,sys,N,T)	generates chains from all N subsets of sys, T=maximum time allowed
chainnormal(sys,chn,vars)	reduces each poly in sys to normal forms using a chain chn
clean(list)	removes a=a or 0 from a list
clearall()	clears all remember tables
clearAssumption()	clears all assumptions made by automated processes in STeM
forgetUnusedEqn()	clears all equations not used by stemnormal during process generation
forgetUsedEqn()	clears all equations used by stemnormal during process generation
comm(oper1,oper2,expr)	computes commutator of oper1,oper2, applied to expression
conj(expr)	computes complex conjugate of expr
convertGHP2NP(expr) ³	converts GHP expression/equation to NP
convertNP2GHP(expr) ³	converts NP expression/equation to GHP
convertNP2Orth(expr) ³	converts NP expression/equation to orthonormal
convertOrth2NP(expr) ³	converts orthonormal expression/equation to NP
defineOperator($Z_{new}, Z_{new} = f(Z_{old})$)	defines new operator in terms of old ones
deglist(list)	applies allddeg to a list
eqns(m,n)	downloads m to n equations of designated formalism
findCombo(sys)	attempts to find a single "suitable" variable combination in sys
findComm(sysde)	finds and computes all possible commutator pairs in sysde
findHalfComm(sysde)	finds and computes all possible higher order commutators in sysde
findVariable(struc,sys,red)	determines possible variable changes, based on user given structure, "struc", which minimises sys according to Maple nops
findops(list)	determines elements of a list of polys which have multiple ops
forgetAssumption()	forces STeM to forget all assumptions
forgetCommutator()	forces STeM to forget computed commutators
formalism(form)	sets the formalism to ghp, np or orth environment
forgetUnusedEqn()	forces STeM to clear all unused equations generated by stemnormal
forgetUsedEqn()	forces STeM to clear all used equations generated by stemnormal
formalism(form)	sets the formalism to ghp, np or orth environment
genreslist(vars,nstrt,eqax,resmax,sys) ⁶	computes all resultants, in pairs, w.r.t. vars of sys

getAssumption()	displays all assumptions made by certain routines used
getUnusedEqn()	retrieves all unused equations generated by automated processes
getUnusedFactor()	retrieves all unused factors generated by automated processes
getUsedEqn()	retrieves all used equations generated by automated processes
getUsedFactor()	retrieves all used factors generated by automated processes
groebcombo(PB,sys,N,T)	generates GB's from all N subsets of sys, T=maximum time allowed
groebreduce(expr,PB, GBL)	reduces expr using each element of GBL
indeppol(sys, nfacs, zerofacs)	routine removes nonzero factors (nfacs: list), and dependant polys which have factors 'zerofacs' from sys
init()	clears all computation results and definitions in session
isCommutable(expr)	checks if a particular expression is commutable ⁵
linreslist(vars,num1,maxl,maxres,list)	computes resultants (length< maxres) wrt linear vars[i],starting from num1 to maximum length maxl of polys in list
minimisePoly(sys)	minimises, in pairs of expr/eqns, Maple length of sys using GB
nullRotation(expr/eqn,preserve=l,n,or ln) ³	performs null rotation preserving l, n, or l,n on expr/eqn
orthRotation(expr/eqn, preserve=e _i) ³	performs 2-d rotation of type preserving e _i , on expr/eqn
picfac(expr)	selects a factor choice from ops of expr
polydecomp(poly,facexp)	decompose poly using factor facexp
polyiso(expr,combo)	isolates a combo from expr
powsubs2	allows multiple powsubs substitutions
prime(expr/eqn)	computes the prime of an expr/equation
properties(var)	displays zero/nonzero/complex/real/weight/prime/star of var
reduceMin(eqn,sys,keep=select) ⁹	reduces the Maple length of eqn w.r.t. sys using LCM
reduceMinSys(sys,keep=select) ⁹	reduces the Maple length of sys w.r.t. itself using term-by-term LCM
remdeg(sys,var,N)	removes all polys from sys with degree > N in var
remfacs(sys,[fac1,fac2,..])	removes factors fac1,fac2,.. from each poly of sys
removeNZfactor(expr)	removes known nonzero factors as given in showAssumption, from expr
remredun(sys)	removes redundant (by numerical factors) polys from sys(polys)
replaceOperator(Z _{new} , Z _{new} = f(Z _{old}))	replaces old operator with defined new one
separate(list)	separates polys from des from a list
sepdeg(var,N,L)	selects polys of L with $0 < \deg(\text{var}) \leq N$
sepeqns(L,varin,varout)	selects polys of L with variable varin but not varout
showAssumption()	displays cumulatively all nonzero assumptions made by automated processes
showCommutator()	shows all commutators calculated by findComm

showEInfo(poly)	displays information of poly
showStatistic(sys)	displays the statistics of sys
showUnusedEqnStat()	displays statistics of unused eqns by stemnormal
showUnusedFactorStat()	displays statistics of unused factors by stemnormal
simplifySys(lista,listb)	minimises pairwise, length of each element of lista against, listb using GB ⁴
specs(sys)	displays indets, length, nops and type of sys
specsop(sys)	displays indets, length, nops and type of each op of sys
star(expr/eqn)	applies the star operation
stemCollect(expr,vars)	collects an expr w.r.t.vars list
stemHTerm(poly)	displays highest ordered term in poly
stemassume([expr1, expr2, ...]) ⁵	declares expr1, expr2,... as nonzero for automated processes
stemcomplete(sys,switch=[op1,op2,op3]) ⁸	proceeds in automated fashion, continuous computation of commutators and subsequently adjoins results to sys then normalises Stops when no new commutators can be found and system is fully normalised/reduced (see stemnormal)
stemdefine(var,weight=[p,q],prime=var1,star=var2)	defines the weight, prime and star of var in a session
stemdefoper([Z1,Z2,Z3,Z4]):	defines an ordering on the operators, from lowest to highest
stemdeforder([var1, var2,...])	defines an ordering on the variables, from lowest to highest
stemeval(expr)	same as maple eval plus complex quantity evaluation in np/ghp
steminvariant([il]) invsys=prime, star or both	establishes an environment automatically invariant under invsys imposing conditions that are invariant as indicated by type,invariant
stemiso(eqn)	solves for derivatives in an equation
stemisoh(eqn/expr)	solves eqn/expr for the highest ordered term,
stemnormal(sys,switch=[op1,op2,op3])	transforms a given system of equations into a normal form based on an ordering defined by stemdeforder and stemdefoper
stemreduce(eqn/expr,sysn)	reduces, a given eqn/expr w.r.t. sysn ¹
stemsorte(sys)	sorts a system, in a solved form with highest term on LHS, in ascending order with all polynomials listed first
stemundefine(var)	clears all defined properties of var.
subtractpair (sys)	subtracts and reduces in pairs of sys.
symtran(sys,nvars,symnvars)	performs discrete symmetry transformation on sys ⁷
weight(expr/eqn/var)	displays the weight of a given expr/eqn/var

In the tables below, we indicate where the various equation groups of the basic formalisms, are located.

Newman-Penrose Equation System

Position	Description
1..18	Ricci
19..29	Bianchi
30..33	Maxwell
34..45	Killing
46..57	First IC Killing ²
58..72	Second IC Killing ²

Ellis-MacCallum Equation System

Position	Description (Eqns)
1..16	Jacobi
17..26	Field
27..42	Bianchi
43..48	E_{ab}
49..54	H_{ab}
55..58	Contracted Bianchi
59..66	Maxwell
67..76	Killing
77..81	$E_{ab;c}u^c$
82..86	$H_{ab;c}u^c$
87..104	Basic Variables
105..135	Basic EFE/I.C.
136..148	Ext./New Variables ¹⁰
149..153	λ and $e_i(\lambda)$

Geroch-Held-Penrose Equation System

Position	Description (Eqns)
1..12	Ricci
13..24	Bianchi
25..28	Maxwell
29..40	Killing
41..52	First IC Killing ²
53..67	Second IC Killing ²

In the above tables we have used the following abbreviations: (i). sys=list of expressions/equations of polynomials; (ii). expr=expression; (iii). Z1,Z2,..denote operators; (iv). x1,x2,..denote variables; (v). vars is a variables list, var is one variable (vi) expr is an expression with variables and/or operators; (vii). form=np,orth, or ghp; (viii) R= Polynomial Ring (ix). N=integer. (x) PB list of polynomials for ideal (xi) GBL: Groebner basis list.(xii): zerofacs: factors that are zero.¹The given system of differential equations, *sysdeqn*, is required to be in a normal form, as given by *stemnormal*. Warning: if the given system is not in a normal form, this function might get into an infinite loop;² The integrability conditions are only for a Homothetic vector;³ The *STeM* environment for the rotations/conversions must be free of global assignments, i.e. in its generic state with original variables;⁴Groebner Bases; ⁵An expression is potentially commutable if when it is solved for the highest ordered derivative all other derivatives on r.h.s. are of lower order;⁶nstrt: poly number in sys to begin resultants; eqmax: maximum length of polys used; resmax: maximum size of resultant to retain;⁷nvars: new extra variables introduced and symnvars the associated symmetry mapping of nvars.⁸Optional entries are: "algebraic", "factor" and "immstop" for control of substitution of algebraic conditions, factors, and the terminate option after each decision step. The user should exercising care when "cutting" output from *STeM* and then pasting it in as input, in the orthonormal formalism. There is no issue with the variables but the operators paste as "e[n]", n=0..3, which is not acceptable. These may be manually changed to "en" or done so through the use of the routine convder. ⁹The switch "select=keep" gives the user the option of which factors to keep in the reduction process otherwise automatically the largest ordered/size factor is retained. ¹⁰ $\omega=w3=-O3$ in the aligned frame

STeM running under Maple, contains all of the standard equations of the Orthonormal, NP and GHP formalisms. It automates virtually all of the regular computational tasks present in the tetrad frames, like propagation, commutation and normalisation of differential/polynomials conditions. The main key routine in *STeM* is *stemnormal* which reduces a combined list/set of polynomial and differential conditions to a "standard" form with the order of the variables and operators as specified by the user. In addition, there are five major simplifying routines in *STeM* called *stemreduce*, *minimisePoly*, *simplifySys*, *reduceMinSys* and *stemHTerm* to minimize/simplify resulting systems of equations. Briefly, *stemreduce* reduces a list of polynomial/differential conditions to a "minimal form" with respect to another list of conditions. *minimisePoly* is a routine which attempts to simplify, in pairs, elements of a set of polynomial conditions using Groebner basis methods. It can readily handle systems of large polynomial equations. *simplifySys* applies *minimisePoly* to two separate lists

minimizing/simplifying one against the other. *reduceMinSys* (*reduceMin* is for the reduction of a single condition with respect to a list of conditions) is a routine which performs a sort of exhaustive pair-term elimination using LCM's, to minimise the Maple length (or number of terms). It is computationally very intense and is more suited to small or medium sized systems. Large systems can also be addressed but would require considerably more time to complete the normalization/reduction process. *stemHTerm* presents the leading terms in a set of polynomial, based on the user's defined variable order, and then other routines in *STeM* decide, based on these leading terms, what equations are pairwise "suitable" for elimination of leading terms and hence is a sort of "restricted" Groebner method. Again, the governing principle is based on minimising Maple length, or variable elimination or term-number reduction in polynomials. Combined with the use of standard resultants, these tools can be quite effective in reducing/simplifying polynomial systems.

In the next section, we will present three illustrative applications showing how *STeM* facilitates the computations typically encountered in the main tetrad formalisms. Example 1 concerning shear-free dust, is the simplest and is at an introductory level. The second example is at an intermediate level requiring more sophisticated use of Maple and *STeM* commands. The final example is for the advanced user who is quite proficient in Maple and demonstrates some of the most conceptually complex applications of *STeM* commands¹

4 *STeM* Applications

In the applications that follow, we shall present the exact Maple input and some of the output that is required to achieve a particular result. For cases where intermediate steps may be large, we shall suppress the output print. The key feature that we want to emphasize is that in many problems intermediate expressions may be very large and it would be of little practical value to explicitly see them. Instead, the processes and advanced tools available in *STeM* allow the user to properly and intelligently navigate through a particular sequence of very large intermediate steps using only *STeM* auxiliary information, provided by *showStatistic*, *specs*, *maplen*, *sortl*, and other routines. Of course there are also generic Maple internal commands that can be used as well

4.1 Example 1: Shearfree dust spacetimes

We will now illustrate the use of some of *STeM*'s routines with a simple shearfree, dust spacetime, using the orthonormal formalism. We begin by reading in the *STeM* package and designate the formalism we wish to use which in this case is the orthonormal one. This is ideal for the study of perfect fluid spacetimes. Notice, that in addition, we will also use the routines in Maple's *student* package,

```
>read 'c:/STeMV1R1.mpl';
```

STeM_v1r1 (June-2026)(c), by K. T. Vu and J. Carminati

The authors would kindly request that this software be referenced if it is used in work eventuating in a publication

Users are free to modify this program but it is not to be redistributed in modified form

We use the Maple command *with()*, to load the *STeM* package:

```
>with(STeM);
```

```
[D, E, EP, T, TP, V, X, Y, addpair, alldg, autocomm, buildchain, chaincombo, chainnormal, clean, clearAssumption,
clearUnusedEqn, clearUsedEqn, clearall, comm, conj, convder, convertGHP2NP, convertNP2GHP, convertNP2Orth,
convertOrth2NP, defineOperator, deglist, e0, e1, e2, e3, eqns, findCombo, findComm, findHalfComm, findVariable,
findops, forgetAssumption, forgetCommutator, forgetUnusedEqn, forgetUsedEqn, forgetUsedFactor, formalism,
genreslist, genresproc, getAssumption, getUnusedEqn, getUnusedFactor, getUsedEqn, getUsedFactor, grobeqn,
grobred, groebcombo, groebreduce, indeppol, init, isCommutable, linreslist, minimisePoly, nullRotation, orthRotation,
picfac, polydecomp, polyiso, powsubs2, prime, properties, reduceMin, reduceMinSys, remdeg, remdeppol, remeqns,
remfac, removeNZfactor, remredun, replaceOperator, resproc, rmsqr, separate, sepdeg, sepeqns, showAssumption,
showCommutator, showEInfo, showStatistic, showUnusedEqnStat, showUnusedFactorStat, showUsedEqnStat,
showUsedFactorStat, simplifySys, specres, specs, specsop, star, stemCollect, stemConvert, stemHTerm, stemassume,
stemcomplete, stemdefine, stemdefoper, stemdeforder, stemeval, steminvariant, stemiso, stemisoh, stemnormal,
stemreduce, stemsorte, stemundefine, subtractpair, symtran, tranfT, weight]
```

The *STeM* command *formalism()* is used to specify which formalism is to be used in a Maple worksheet. It is recommended that this command to be the first issued *STeM* command. In this example, we shall use the orthonormal formalism.

```
>formalism(orth);
```

At this stage, it is important to define an ordering on the variables and the operators which are present. We use the *STeM* command *stemdeforder()* to specify the order of the variables (from low to high).

```
>stemdeforder([M,H,omega,z1,z2,z3,q1,q2,q3,r1,r2,r3,n,n33,E11,E22,E12,E13,E23,M11,M22,M12,M13,M23,j]):
```

The *STeM* command *stemdeforder()* specifies the order of the operators (from low to high).

```
>stemdefoper([e3,e2,e1,e0]):
```

We state any assumptions inherent in the analysis, by using the *STeM* command *stemassume()*. Note that it provides the argument as a list or set. For this specific example, we wish to impose nonzero assumptions on the physical variables, *M* the energy density, *H* the fluid expansion and *omega* the fluid rotation rate scalar

```
> stemassume(M,omega,H):
```

The mathematical conditions representing our shearfree, rotating, expanding dust model are introduced by making the following global assumptions on σ_{ij} , the shear tensor components, $\langle w1, w2, w3 \rangle$ the fluid vorticity vector and $\langle O1, O2, O3 \rangle$ the tetrad rotation vector.

```
>theta:=H:E33:=-E22-E11:M33:=-M22-M11:H1:=H/3:H2:=H/3:H3:=H/3:s12:=0:s23:=0:s13:=0::u1:=0:u2:=0:u3:=0:
```

```
>w1:=0:w2:=0:w3:=omega:O1:=0:O2:=0:O3:=omega:n22:=n:n11:=n:P:=0:
```

We note that a certain gauge freedom has been used in order to fully establish all these conditions. Before we begin, it is useful to make the following simple global variable change, as shown. We use the *STeM* command *eqns()* to generate the appropriate equations, which are located in positions 140 to 145.

```
>new_vars:=eqns(140,145);assign(%);
```

```
new_vars:=[a1 = (1/2)r1-(1/2)q1, a2 = (1/2)r2-(1/2)q2, a3 = (1/2)r3-(1/2)q3, n12 = (1/2)r3+(1/2)q3, n13 =
(1/2)r2+(1/2)q2, n23 = (1/2)r1+(1/2)q1]
```

Next, the Maple command *assign()* is used to define *a1,a2,a3, n12,n13,n23* globally.

```
>assign(new_vars):
```

It will prove useful to introduce the extension variables $\{z1, z2, z3, j\}$ which are located in positions 136 to 139 and are displayed as follows.

```
>ext_var:=eqns(136,139);
```

```
ext_var:=[e1(H) = z1, e2(H) = z2, e3(H) = z3, e0(H) = -(1/3)H^2+2*omega^2-(1/2)M-(3/2)P+j]
```

Finally, we complete the preparation stage by generating the complete set of field equations, positioned from 1 to 58, using *STeM*'s *eqns* command.

```
>field_eqs:=eqns(1,58):
```

We initiate the main phase of the computation by using *STeM*'s *stemnormal()* command to normalise/reduce the system of equations composed of *ext_var* and *field_eqs*. The options *switch=[factor,algebraic,immstop]* may be used to provide the user with additional decision choices. For this example, we will leave these options out. This process will bring the system into its simplest standard form with the given orderings.

```
>hqa:=stemnormal([op(ext_var),op(field_eqs)]);
```

```
hqa := [n33 = 2z3/(3*omega), E11 = (1/3)*omega^2, E22 = (1/3)*omega^2, E12 = 0, E13 = 0, E23 = 0, M11 = -omega*r3, M22 = omega*q3,
M12 = 0, M13 = -omega*q1+(1/3)*z2, M23 = omega*r2-(1/3)*z1, j = 0, e3(M) = -8*omega^2*(q3-r3), e2(M) = (1/3)*omega*(24*omega*r2-13*z1),
e1(M) = -(1/3)*omega*(24*omega*q1-13*z2), e0(M) = -HM, e3(H) = z3, e2(H) = z2, e1(H) = z1, e0(H) = -(1/3)H^2+2*omega^2-(1/2)M,
e3(omega) = -omega*(q3-r3), e2(omega) = omega*r2-2z1/(3*omega), e1(omega) = -omega*q1+2z2/(3*omega), e0(omega) = -2H*omega/(3*omega), e2(z1) = (-2H^2*omega^2+12*omega^4-
3M*omega^2-6*omega*r1z2-2z3^2)/(6*omega), e0(z1) = -Hz1+(1/2)*omega*z2, e2(z2) = q2z2-z3q3+r1z1-z3r3+e1(z1), e1(z2) = -(-2H^2*omega^2+
12*omega^4-3M*omega^2-6*omega*q2z1-2z3^2)/(6*omega), e0(z2) = -Hz2-(1/2)*omega*z1, e3(z3) = 2H*omega^2/(3*omega)-q1z1+r2z2-q2z2+z3r3-e1(z1),
e2(z3) = (3*omega*q3z2+z1z3)/(3*omega), e1(z3) = (-3*omega*r3z1-z2z3)/(3*omega), e0(z3) = -z3H, e2(q1) = (H*omega^2+9*q3n*omega+9*r3n*omega+
9*q1r2*omega+9*r1r2*omega+3*q1z1-9*z3q3-3r2z2+3*q2z2+3*e1(z1))/(9*omega), e0(q1) = -(1/3)H*q1+(1/3)*z1, e0(q2) = -(1/3)H*q2+
(1/3)*z2, e1(q3) = (3*omega^2*q3r1+3*omega^2*r1r3-omega*q3z2+3*omega*r2z3-z1z3)/(3*omega^2), e0(q3) = -(1/3)H*q3+(1/3)*z3, e1(r1) = (9*r1^2*omega^2+
3M*omega^2+9*e2(q2)*omega^2+3z3^2-H^2*omega^2+9*q2^2*omega^2-9*r3q3*omega^2-6*nz3*omega-21*omega^4)/(9*omega^2), e0(r1) = -(1/3)H*r1-(1/3)*z1,
e3(r2) = (9*nq1*omega^2-9*q2q3*omega^2-9*q2r3*omega^2-9*r2r3*omega^2-9*e2(q3)*omega^2-3*nz2*omega-6*q1z3*omega+12*q3z1*omega-3*r3z1*omega+3*e3(z1)*omega+2z3z2)/(9*omega^2),
e2(r2) = (-H^2*omega^2-3*omega^4-9*r1q1*omega^2+9*q3^2*omega^2+9*r2^2*omega^2+3M*omega^2+9*e3(q3)*omega^2-z3^2)/(9*omega^2), e1(r2) = (H*omega^2-9*q3n*omega-9*r3n*omega-9*q1q2*omega-
9*q1r2*omega+3*q1z1+3*q2z2-3z3q3-3r2z2+6z3r3+3*e1(z1))/(9*omega), e0(r2) = -(1/3)H*r2-(1/3)*z2,
e3(r3) = (-3*omega^4+9*q1^2*omega^2-9*r2q2*omega^2+9*r3^2*omega^2+3M*omega^2-z3^2-H^2*omega^2+9*e1(q1)*omega^2)/(9*omega^2), e2(r3) = (-3*omega^2*q2q3-3*omega^2*q2r3-
3*omega*q1z3+omega*r3z1+z2z3)/(3*omega^2), e1(r3) = (-9*nz2*omega^2+9*q1q3*omega^2+9*r1q3*omega^2+9*r1r3*omega^2-9*e3(q1)*omega^2+3*nz1*omega+3*q3z2*omega+6*r2z3*omega-
12*r3z2*omega+3*e3(z2)*omega-2z3z1)/(9*omega^2), e0(r3) = -(1/3)H*r3-(1/3)*z3, e2(n) = (-3*nq2*omega^2+3*nz2*omega^2-3*q1q3*omega^2-3*r1q3*omega^2-
3*e3(r1)*omega^2-3r2z3*omega+q3z2*omega+z3z1)/(3*omega^2), e1(n) = (-3*nq1*omega^2+3*nz1*omega^2+3*q2r3*omega^2+3r2r3*omega^2-3*e3(q2)*omega^2+3*q1z3*omega-r3z1*omega-
z3z2)/(3*omega^2), e0(n) = -(1/3)nH]
```

For the above computation, we use *STeM*'s *showAssumption()* to view all assumptions made.

```
>showAssumption();
```

Non-zero assumption predefined by user

$H \neq 0$

$M \neq 0$

$\omega \neq 0$

The command *showStatistic()*, can provide us with the full details of the system hqa.

```
>showStatistic(hqa);
```

```
Number of equations are , 51
Number of polynomial equations are , 12
Number of differential equations are , 39
```

```
Equations' length are
```

```
[28, 23, 23, 6, 6, 6, 24, 20, 6, 33, 33, 4, 38, 56, 56, 24, 13, 13, 13, 44, 42, 42, 42, 36, 109, 47, 81, 109, 47, 101, 69, 69,
26, 203, 39, 39, 135, 39, 188, 39, 270, 166, 217, 39, 166, 135, 270, 39, 200, 200, 28]
```

```
Equations' LHS are
```

```
[n33, E11, E22, E12, E13, E23, M11, M22, M12, M13, M23, j, e3(M), e2(M), e1(M), e0(M), e3(H), e2(H), e1(H),
e0(H), e3(ω), e2(ω), e1(ω), e0(ω), e2(z1), e0(z1), e2(z2), e1(z2), e0(z2), e3(z3), e2(z3), e1(z3), e0(z3), e2(q1), e0(q1),
e0(q2), e1(q3), e0(q3), e1(r1), e0(r1), e3(r2), e2(r2), e1(r2), e0(r2), e3(r3), e2(r3), e1(r3), e0(r3), e2(n), e1(n), e0(n)]
```

We see that the system hqa has 51 equations in total, with 12 conditions being polynomials and 39 being differential. The first list is composed of the Maple length of each equation in hqa. The second list shows the leading term (the one that is solved for) of each equation, corresponding to the ordering as given initially by the user. Next, we will demonstrate how integrability conditions, through the use of commutators can be derived from a given system. For example, if we wish to obtain the integrability condition on M resulting from the $[e_0, e_1]M$ commutator, then we would use *STEM's comm()* command as follows.

```
>comm_M:=comm(e0,e1,M);
```

```
comm_M := e0(e1(M))-e1(e0(M)) = -(1/3)*H*e1(M)
```

Next, we can isolate (bringing to the left-hand-side) the highest ordered derivative, in this condition, using *STEM's stemisoh()*:

```
comm_Ma:=stemisoh(comm_M);
```

```
comm_Ma := e1(e0(M)) = e0(e1(M))+(1/3)He1(M).
```

We can reduce a commutator against a system, say hqa, which was generated above, by using *STEM's stemreduce()*.

```
>stemreduce(comm_M,hqa);
```

```
(48Hq1-87z1)ω2-26Hωz2+18z1M
```

Notice, that the commutator has been reduced to a useful polynomial condition. This is a key ingredient in the analysis of shearfree dust spacetimes. If the user wishes to generate all commutators on a variable, say M, then this can be achieved with the use of *autocomm(M)*.

```
>allcomm_M := autocomm(M);
```

```
allcomm_M := [3e0(e1(M))-3e1(e0(M))+He1(M) = 0, 3e0(e2(M))-3e2(e0(M))+He2(M) = 0,
3e0(e3(M))-3e3(e0(M))+He3(M) = 0, e1(e2(M))-e2(e1(M))+2ωe0(M)-q2e1(M)-r1e2(M)-n33e3(M) = 0,
e1(e3(M))-e3(e1(M))+r3e1(M)+ne2(M)+q1e3(M) = 0, e2(e3(M))-e3(e2(M))-ne1(M)-q3e2(M)-r2e3(M) = 0]
```

Now, with any given general differential system, *sys*, not all 6 commutators on all the variables present can be reduced with respect to *sys*. Interestingly, *STEM's* routine *findComm()*, can actually, intelligently determine which commutators can be reduced (can be performed or are possible) and subsequently carry out the reduction process. We shall now apply this routine, combined with *stemsorte()* which sorts the equations from low to high order, to hqa .

```
>totcomm:=findComm(hqa);
```

```
>totcomm_a := stemsorte(totcomm);
```

```
totcomm_a := [q1 = -(26Hωz2-87ω2z1+18Mz1)/(48Hω2), q3 = 3z3z2/(ωz1), r2 = (26Hωz1-87ω2z2+18Mz2)/(48Hω2),
r3 = 3z3z1/(ωz2), r3 = (8Hω2q3-16ω2z3+3Mz3)/(8Hω2), e3(z1) = -(3nωz2-3ωq1z3+z2z3)/(3ω),
e3(z1) = (3nωz2+3ωq1z3-z2z3)/(3ω), e3(z1) = -(15nωz2-24ωq1z3-9ωr3z1+8z2z3)/(15ω),
e1(z1) = -5Hω2(1/3)-q2z2+z3r3, e1(z1) = (8/15)Hω2-(3/5)HM-(3/10)q1z1-q2z2-(3/10)z3q3+(3/10)r2z2+(13/10)z3r3,
e1(z1) = (-6H2ω2q3+6H2ω2r3+36ω4q3-36ω4r3+44Hω2z3-9Mω2q3+9Mω2r3-24q2z2z3+24r3z32)/(24z3),
e1(z1) = -(16Hω3-3HMω-3ωq1z1+3ωq2z2+10z1z2)/(3ω), e1(z1) = -(16Hω3-3HMω+3ωq2z2-3ωq3z3+3ωr2z2-3ωr3z3-
10z1z2)/(3ω), e3(z2) = (-3nωz1-3ωr2z3+z1z3)/(3ω), e3(z2) = (-15nωz1-9ωq3z2-24ωr2z3+8z1z3)/(15ω)]
```

We can find out which commutators were actually carried out by *findComm* using *STEM's showCommutator()*.

```
>ShowCommutator();
```

Number of full commutators calculated is , 25

They are

$$\begin{aligned} &[e1, e0, H], [e2, e0, H], [e2, e1, H], [e3, e0, H], [e3, e1, H], [e3, e2, H] \\ &[e1, e0, M], [e2, e0, M], [e2, e1, M], [e3, e0, M], [e3, e1, M], [e3, e2, M] \\ &[e1, e0, \omega], [e2, e0, \omega], [e2, e1, \omega], [e3, e0, \omega], [e3, e1, \omega], [e3, e2, \omega] \\ &[e1, e0, q3] \\ &[e1, e0, z2] \\ &[e1, e0, z3], [e2, e0, z3], [e2, e1, z3] \\ &[e2, e0, r3] \\ &[e2, e0, z1] \end{aligned}$$

Number of half commutators calculated is , 0

Finally, we combine the results from the commutators, totcomm_a, with our system hqa and then normalise the total system, with a switch option, as follows.

> hqb := stemnormal([op(hqa), op(totcomm_a)], switch = [algebraic, factor, immstop]);

"Obtain algebraic equation"

$$24*H*\omega*z1^2-24*H*\omega*z2^2+16*\omega^2*z1*z2-3*M*z1*z2$$

Substitute this algebraic equation into the system? (y,n,ter) n

"Obtain algebraic equation"

$$M11*z2+3*z1*z3$$

Substitute this algebraic equation into the system? (y,n,ter) n

"Obtain algebraic equation"

$$M22*z1-3*z2*z3$$

Substitute this algebraic equation into the system? (y,n,ter) n

"Obtain algebraic equation"

$$48*H*M13*\omega+10*H*\omega*z2+87*\omega^2*z1-18*M*z1$$

Substitute this algebraic equation into the system? (y,n,ter) n

"Obtain algebraic equation"

$$48*H*M23*\omega-10*H*\omega*z1+87*\omega^2*z2-18*M*z2$$

Substitute this algebraic equation into the system? (y,n,ter) n

"Obtain algebraic equation"

$$144*H*\omega*z1^2+10*H*\omega*z2^2+87*\omega^2*z1*z2-18*M*z1*z2$$

Substitute this algebraic equation into the system? (y,n,ter) n

"Obtain algebraic equation"

$$\begin{aligned} &-176*H^2*\omega^4*z2+48*H^2*M*\omega^2*z2-134*H*\omega*z1*z2^2-144*H*\omega*z1*z3^2+87*\omega^2*z1^2*z2- \\ &18*M*z1^2*z2 \end{aligned}$$

Substitute this algebraic equation into the system? (y,n,ter) n

"Obtain algebraic equation"

$$6*H^2*z1^2-6*H^2*z2^2+28*H*\omega*z1*z2-36*\omega^2*z1^2+36*\omega^2*z2^2+9*M*z1^2-9*M*z2^2$$

Substitute this algebraic equation into the system? (y,n,ter) n

"Obtain algebraic equation"

$$\begin{aligned} &-176*H^2*\omega^4*z1+48*H^2*M*\omega^2*z1+134*H*\omega*z1^2*z2+144*H*\omega*z2^2*z3^2+87*\omega^2*z1^2*z2^2- \\ &18*M*z1^2*z2^2 \end{aligned}$$

Substitute this algebraic equation into the system? (y,n,ter) n
 "Obtain algebraic equation"

$$-352^*H^2*\omega^4*z1*z2+96^*H^2*M*\omega^2*z1*z2-144^*H*\omega^3*z1^2*z3^2+144^*H*\omega^3*z2^2*z3^2+87^*\omega^2*z1^3*z2+87^*\omega^2*z1^2*z2^3-18^*M*z1^3*z2-18^*M*z1^2*z2^3$$

Substitute this algebraic equation into the system? (y,n,ter) n
 "Obtain algebraic equation"

$$10^*H*\omega^3*z1^2+144^*H*\omega^3*z2^2-87^*\omega^2*z1*z2+18^*M*z1^3*z2$$

Substitute this algebraic equation into the system? (y,n,ter) n
 Obtain algebraic equation with length = 576
 and containing {H, M, n, omega, z1, z2, z3}
 Substitute this algebraic equation into the system? (y,n,ter) n
 "Obtain algebraic equation"

$$116^*H^2*\omega^2*z1-87^*H*\omega^3*z2+348^*\omega^4*z1+18^*H*M*\omega^3*z2-159^*M*\omega^2*z1+18^*M^2*z1$$

Substitute this algebraic equation into the system? (y,n,ter) n
 "Obtain algebraic equation"

$$24^*H^3*\omega^3*z1*z2+120^*H^2*\omega^4*z2^2-144^*H*\omega^5*z1*z2+36^*H*M*\omega^3*z1*z2+72^*H*\omega^2*q2*z1^2*z2+72^*H*\omega^2*q2*z2^3-72^*H*\omega^2*r1*z1^3-72^*H*\omega^2*r1*z1^2*z2^2-221^*H*\omega^3*z1^3*z2-9^*H*\omega^3*z1^2*z2^3-192^*H*\omega^3*z1^2*z2^3^2+174^*\omega^2*z1^2*z2^2-36^*M*z1^2*z2^2^2$$

Substitute this algebraic equation into the system? (y,n,ter) n
 "Obtain algebraic equation"

$$11^*z1^2+9^*z2^2$$

Substitute this algebraic equation into the system? (y,n,ter) ter

>showAssumption();

Non-zero assumption predefined by user

H ≠ 0
 M ≠ 0
 ω ≠ 0

Non-zero denominators

z1 ≠ 0
 z2 ≠ 0
 z3 ≠ 0

The used conditions are placed in the global set *getUsedEqn()* whereas the unused ones are placed in *getUnusedEqn()*. For example, we can display the unused equations as follows.

>getUnusedEqn();

$$\{M11z2+3z1z3, M22z1-3z2z3, 10H\omega z1^2+144H\omega z2^2-87\omega^2 z1z2+18Mz1z2, 24H\omega z1^2-24H\omega z2^2+16\omega^2 z1z2-3Mz1z2, 144H\omega z1^2+10H\omega z2^2+87\omega^2 z1z2-18Mz1z2, 48HM13\omega+10H\omega z2+87\omega^2 z1-18Mz1, 48HM23\omega-10H\omega z1+87\omega^2 z2-18Mz2, 116H^2\omega^2 z1-87H\omega^3 z2+348\omega^4 z1+18HM\omega z2-159M\omega^2 z1+18M^2 z1, -176H^2\omega^4 z2+48H^2M\omega^2 z2-134H\omega z1z2^2-144H\omega z1z3^2+87\omega^2 z1^2 z2-18Mz1^2 z2, -176H^2\omega^4 z1+48H^2M\omega^2 z1+134H\omega z1^2 z2+144H\omega z2z3^2+87\omega^2 z1z2^2-18Mz1z2^2, 6H^2 z1^2-6H^2 z2^2+28H\omega z1z2-36\omega^2 z1^2+36\omega^2 z2^2+9Mz1^2-9Mz2^2, -352H^2\omega^4 z1z2+96H^2M\omega^2 z1z2-144H\omega z1^2 z3^2+144H\omega z2^2 z3^2+87\omega^2 z1^3 z2+87\omega^2 z1z2^3-18Mz1^3 z2-18Mz1z2^3, 24H^3\omega^3 z1z2+120H^2\omega^4 z2^2-144H\omega^5 z1z2+36HM\omega^3 z1z2+72H\omega^2 q2 z1^2 z2+72H\omega^2 q2 z2^3-72H\omega^2 r1 z1^3-72H\omega^2 r1 z1z2^2-221H\omega z1^3 z2-9H\omega z1z2^3-192H\omega z1z2z3^2+174\omega^2 z1^2 z2^2-36Mz1^2 z2^2,-2448H^3\omega^5 z1z2+8352H\omega^7 z1z2+288H^3M\omega^3 z1z2-6912H^2n\omega^3 z1^2 z3-6912H^2n\omega^3 z2^2 z3-3816HM\omega^5 z1z2-520H^2\omega^2 z1^2 z2^2-2304H^2\omega^2 z1^2 z3^2+3168H^2\omega^2 z2^2 z3^2+432HM^2\omega^3 z1z2-3654H\omega^3 z1^3 z2+3132H\omega^3 z1z2^3-1392H\omega^3 z1z2z3^2+15921\omega^4 z1^2 z2^2+540HM\omega z1^3 z2-648HM\omega z1z2^3+288HM\omega z1z2z3^2-7992M\omega^2 z1^2 z2^2+972M^2 z1^2 z2^2\}.$$

Notice that we artificially stopped the computation because we recognise that we have a positive definite condition so that $z_1=z_2=0$ which contradicts the original assumptions made. What this contradiction states is that the generic case, is incompatible with the user assumption of shear-free dust. Further analysis, for the special cases as dictated by the nonzero factors/denominators will eventually lead to similar contradictions. It is well known that for all such dust spacetimes, the expansion or vorticity must vanish i.e. $H\omega=0$. This is part of a long standing conjecture that general relativistic, shear-free, perfect fluids which obey a barotropic equation of state $P=P(M)$ such that $M+P\neq 0$ must be either nonexpanding or nonrotating. At this point in time this conjecture remains unresolved.

Drawing on considerable experience using *STeM*, we have observed that, given an initial complete set of equations, the resulting intermediate and final states of the system can depend significantly on the ordering of the variables or operators chosen at the outset. Therefore, we would suggest that the user do a little "experimentation" on different orderings to investigate which one produces the most compact final systems, if that is the desired objective. As is evident from the input/output, at each step, when a polynomial condition is generated/found, the user is prompted as to whether or not that condition is to be substituted back into the system or terminate the execution or to continue with the computation without substitution. The question of which conditions to substitute back into the normalizing processes and which ones to keep aside has no definitive answer. There are no hard-and-fast rules here. Depending on the nature of the problem, individual users must decide, perhaps guided by some experimentation or "intuition", on how large a system they wish to retain. Without exercising care at critical points, the computer may be overwhelmed with intermediate swell problems coupled with unacceptable time/memory requirements.

4.2 Example 2: Type D vacuum spacetimes

In this example, we shall examine vacuum Petrov type D metrics, which are solutions of Einstein's field equations, all of which have been previously given by Kinnersley. However, it was some time later that it was shown by Czapor and McLenaghan² that all of these spacetimes possess the intriguing spin coefficient conditions $\tau\bar{\tau} = \pi\bar{\pi}$ and $\rho\bar{\mu} = \bar{\rho}\mu$, in the null frame that is aligned with the principal null directions of the Weyl tensor. In their computation using the Maple program NP,² these identities were established, though with considerable difficulty. They used the NP formalism with a special gauge choice of $\rho = \pi$ and then performed lengthy computations with trial-and-error attempts with various substitution/reduction steps. Clearly, this can be a difficult computational problem. We shall show, using *STeM*, how these identities can be established in a simple, virtually automated fashion and in just a few minutes. Almost all of the input and output are displayed below.

We begin with the *STeM* input to establish the appropriate GHP environment with conditions and equations

```
>read 'c:/STeMV1R1.mpl':with(STeM):formalism(ghp):eqs:=eqns(1,24):
>stemdeforder([m,p,r,t,W2]):stemdefoper([E,EP,T,TP]):
>R00:=0:R01:=0:R02:=0:R12:=0:R10:=0:R20:=0:R21:=0:R22:=0:R11:=0:W0:=0:W0_c:=0:W1:=0:
W1_c:=0:W3:=0:W3_c:=0:W4:=0:W4_c:=0:k:=0:k_c:=0:s:=0:s_c:=0:n:=0:n_c:=0:l:=0:l_c:=0:L:=0:
```

A few clarify points need to be made at this stage. Firstly, we need to bear in mind that the conjugate variables are treated as independent from the variables themselves. Secondly, it is generally best to position the curvature at a higher order than the spin coefficients. However, having said that, we have noted that there have been some (very rare) exceptional cases where the best ordering is reversed. The ordering dictates to *stemnormal* (and also other routines) a key routine in *STeM*, which variables have a higher priority for the solving/reduction processes. There is no need to include the conjugates in the ordering as they are automatically positioned adjacent to their corresponding variables. If the user wishes, the conjugates can be repositioned as desired. Finally we have taken advantage of the Goldberg-Sachs theorem which in this cases yields $\kappa = \sigma = \nu = \lambda = 0$. In any case, one need not add these conditions in as they are otherwise quickly obtained by *STeM* in the reduction/normalization process.

Next, we generate the conjugates of all the equations *eqs* adjoin them to *eqs*, and apply *stemnormal* to the combined set.

```
>ceqs:=map(conj,eqs):
>toteqs:=stemnormal([op(eqs),op(ceqs)],switch=[algebraic,factor]):
toteqs := [EP(m) = -(m-m_c)p, TP(m) = -m^2, E(m_c) = p_c(m-m_c), TP(m_c) = -m_c^2, E(p) = T(m)-mr_c-pp_c-W2, EP(p) = -p^2, TP(p) = -(t_c+p)m, E(p_c) = -p_c^2, EP(p_c) = T(m_c)-m_cr-pp_c-W2_c, TP(p_c) = -m_c(t+p_c), E(r) = (r-r_c)t, T(r) = r^2, EP(r_c) = -t_c(r-r_c), T(r_c) = r_c^2, E(t) = t^2, EP(t) = TP(r)+m_cr+tt_c+W2, T(t) = (t+p_c)r, E(t_c) = TP(r_c)+mr_c+tt_c+W2_c, EP(t_c) = t_c^2, T(t_c) = r_c(t_c+p), E(W2) = 3W2t, EP(W2) = -3W2p, T(W2) = 3W2r, TP(W2) = -3W2m, E(W2_c) = -3W2_cp_c, EP(W2_c) = 3W2_ct_c, T(W2_c) = 3W2_cr_c, TP(W2_c) = -3W2_cm_c]
```

We proceed, by computing all commutators initially possible with *toteqs*.

```
>allcom:=findComm(toteqs):
```


allcom := [E(r_c) = -T(p_c), EP(r) = -T(p), TP(r) = -tt_c-T(m)+pp_c, TP(r_c) = -tt_c-T(m_c)+pp_c, TP(t) = -E(m), TP(t_c) = -EP(m_c)]

We can query *STeM* to see what commutators have been done.

```
>showCommutator();
```

Number of full commutators calculated is , 20

They are

[E, EP, W2], [E, T, W2], [E, TP, W2], [EP, T, W2], [EP, TP, W2], [T, TP, W2]
[E, EP, W2_c], [E, T, W2_c], [E, TP, W2_c], [EP, T, W2_c], [EP, TP, W2_c], [T, TP, W2_c]
[E, T, r], [E, T, t], [E, TP, m_c], [E, TP, p_c], [EP, T, r_c], [EP, T, t_c], [EP, TP, m], [EP, TP, p]
Number of half commutators calculated is , 0

We note that the conjugate variable commutators are also done so that the user need not manually add them. There is also the added benefit that it checks that all results are self consistent. This would assist in catching input typing errors, if they occur. Also we have the added information that no higher order commutators, indicated by "half commutators", have been computed. It is important to always check what assumptions *STeM* has made in given computations. This information can be retrieved as follows, and is always cumulative unless cleared by the user using the *clearAssumption* command at any stage.

```
>showAssumption();
```

Non-zero factors

W2 ≠ 0

W2_c ≠ 0

Now it is evident that the results we are seeking do not lie at the first-order commutator level. We need to address the issue of higher order commutators. We can complete this exercise by first applying *stemnormal* to the combined system consisting of *toteqs* and *allcom*.

```
>toteqsa:=stemnormal([op(toteqs),op(allcom)],switch=[algebraic,factor]);
```

toteqsa := [EP(m) = -(m-m_c)p, TP(m) = -m², E(m_c) = p_c(m-m_c), TP(m_c) = -m_c², E(p) = T(m)-mr_c-pp_c-W2, EP(p) = -p², TP(p) = -(t_c+p)m, E(p_c) = -p_c², EP(p_c) = T(m_c)-m_cr-pp_c-W2_c, TP(p_c) = -m_c(t+p_c), E(r) = (r-r_c)t, EP(r) = -T(p), T(r) = r², TP(r) = -tt_c-T(m)+pp_c, E(r_c) = -T(p_c), EP(r_c) = -t_c(r-r_c), T(r_c) = r_c², TP(r_c) = -tt_c-T(m_c)+pp_c, E(t) = t², EP(t) = -T(m)+pp_c+m_cr+W2, T(t) = (t+p_c)r, TP(t) = -E(m), E(t_c) = -T(m_c)+pp_c+mr_c+W2_c, EP(t_c) = t_c², T(t_c) = r_c(t_c+p), TP(t_c) = -EP(m_c), E(W2) = 3W2t, EP(W2) = -3W2p, T(W2) = 3W2r, TP(W2) = -3W2m, E(W2_c) = -3W2_cp_c, EP(W2_c) = 3W2_ct_c, T(W2_c) = 3W2_cr_c, TP(W2_c) = -3W2_cm_c]

We are now in a position to compute the half-commutators, which compute higher order commutators.

```
>halfcomm:=findHalfComm(toteqsa):
```

```
>maplen(halfcomm);
```

{35, 36, 41, 44, 112, 123, 135, 138, 139, 146, 149, 154, 155, 169, 170, 175, 178, 188, 194}

```
>halfcomm[1..5];
```

{-3pT(p)-EP(T(p)), -3p_cT(p_c)-E(T(p_c)), -3tE(m)+E(E(m)), -3t_cEP(m_c)+EP(EP(m_c)), T(T(p))-3rT(p)}

Because of space limitations, here we have not presented the full output. As expected, this routine has now generated expressions for second order derivatives. The final two steps consist of normalizing *halfcomm* with *toteqsa* and then computing the final set of commutators.

```
>toteqsb:=stemsorte(stemnormal([op(halfcomm),op(toteqsa)],switch=[algebraic,factor]));
```

```
>showStatistic(%);
```

Number of equations are , 58

Number of polynomial equations are , 0

Number of differential equations are , 58

Equations' length are

[34, 20, 33, 24, 54, 20, 34, 23, 61, 38, 29, 22, 15, 50, 25, 38, 19, 54, 15, 55, 29, 22, 60, 20, 33, 27, 25, 26, 25, 26, 31, 32, 31, 32, 35, 175, 155, 112, 135, 139, 36, 178, 170, 123, 44, 188, 149, 155, 44, 194, 154, 36, 35, 138, 41, 169, 41, 146]

Equations' LHS are

```
[EP(m), TP(m), E(m_c), TP(m_c), E(p), EP(p), TP(p), E(p_c), EP(p_c), TP(p_c), E(r), EP(r), T(r), TP(r),
E(r_c), EP(r_c), T(r_c), TP(r_c), E(t), EP(t), T(t), TP(t), E(t_c), EP(t_c), T(t_c), TP(t_c), E(W2), EP(W2),
T(W2), TP(W2), E(W2_c), EP(W2_c), T(W2_c), TP(W2_c), E(E(m)), E(T(m)), EP(E(m)), EP(T(m)), T(E(m)),
T(T(m)), TP(E(m)), TP(T(m)), E(EP(m_c)), E(T(m_c)), EP(EP(m_c)), EP(T(m_c)), T(EP(m_c)), T(T(m_c)),
TP(EP(m_c)), TP(T(m_c)), E(T(p)), EP(T(p)), T(T(p)), TP(T(p)), E(T(p_c)), EP(T(p_c)), T(T(p_c)), TP(T(p_c))]
>allcoma:=findComm(toteqsb);
```

```
allcoma := [t_c = (-mr_c+m_cr+pp_c)/t, t_c = -(-mr_c+m_cr+pp_c)/t, E(m) = mp_c+2mt-m_ct EP(m_c) =
-mt_c+m_cp+2m_ct_c, T(p) = 2pr-pr_c+rt_c, T(p_c) = -p_cr+2p_cr_c+r_ct]
```

```
>factor( numer(rhs(-allcoma[1]+allcoma[2])) )/2; factor( numer(oso(allcoma[1]+allcoma[2])) )/2;
```

$$\frac{mr_c - m_{cr}}{-pp_c + tt_c}$$

We have established the identities that we were looking for plus some additional information on the derivatives. Again, we should check for any additional assumptions made by *STeM*.

```
>showAssumption();
```

```
Non-zero factors
W2 ≠ 0
W2_c ≠ 0
Non-zero denominators
t ≠ 0
t_c ≠ 0
```

Note that we have picked up the extra assumption that $\tau \neq 0$. Presumably, to complete the investigation, one should check the $\tau = 0$ subcase. We wish to emphasize that at no stage was there any trial-and-error analysis and that all computations were carried out in an automated fashion. Ultimately, this whole computation can be carried out more efficiently and more rapidly if one uses the *stemcomplete* command as follows; all output suppressed except for the final result

```
>alleqs:=stemnormal([op(eqs),op(ceqs)],switch=[algebraic,factor]):
>alleqsa:=stemcomplete(alleqs):
>halfcomm:=findHalfComm(alleqsa):
>alleqsb:=stemnormal([op(alleqsa),op(halfcomm)],switch=[algebraic,factor]):
>fineq:=stemsorte(stemcomplete(alleqsb)):
```

We can actually know a good deal about the system *fineq*, without explicitly displaying it, by using *showStatistic*. This can be useful for when one is dealing with very large systems of large equations.

```
>showStatistic(fineq);
```

```
Number of equations is , 34
Number of polynomial equations is , 2
Number of differential equations is , 32
Equations' lengths are
[21, 21, 51, 34, 189, 20, 33, 74, 173, 24, 189, 20, 71, 42, 23, 174, 73, 38, 37, 76, 15, 190, 15, 190, 29, 52, 25, 26,
25, 26, 31, 40, 39, 32]
Equations' LHS are
[r_c, t_c, E(m), EP(m), T(m), TP(m), E(m_c), EP(m_c), T(m_c), TP(m_c), E(p), EP(p), T(p), TP(p),
E(p_c), EP(p_c), T(p_c), TP(p_c), E(r), EP(r), T(r), TP(r), E(t), EP(t), T(t), TP(t), E(W2), EP(W2),
T(W2), TP(W2), E(W2_c), EP(W2_c), T(W2_c), TP(W2_c), TP(W2_c)]
```

The total process has determined the desired two polynomial conditions, together with 32 differential conditions, which because of the *stemsorte* command will be located at the beginning of the list of equations *fineq*.

```
>fineq[1];fineq[2];
```

$$\begin{aligned} r_c &= m_{cr}/m \\ t_c &= pp_c/t \end{aligned}$$

However, in this case, the system, through the use of *stemcomplete*, does make more nonzero assumptions. For example *STeM* assumes the following nonzero factors

$$\begin{aligned}
&\text{Non-zero factors} \\
&p \neq 0 \\
&r \neq 0 \\
&W2 \neq 0 \\
&p_c \neq 0 \\
&r_c \neq 0 \\
&W2_c \neq 0 \\
&p + 2\,t_c \neq 0 \\
&p_c + 2\,t \neq 0 \\
&2\,m_c\,p_c\,r + 2\,m_c\,r\,t + 2\,p\,p_c + 2\,p\,p_c\,t + W2\,p_c + W2_c\,t \neq 0 \\
&2\,m\,p\,r_c + 2\,m\,r_c\,t_c + 2\,p\,p_c + 2\,p\,p_c\,t_c + W2\,t_c + W2_c\,p \neq 0
\end{aligned}$$

Similarly for nonzero assumptions made on denominators. In general the price one pays for more automation is the possible introduction of additional assumptions. These special cases need further analysis. As a final point, one can, using *stemassume*, declare that certain quantities are nonzero so that *STeM* carries out the computations assuming they are not branching decisions.

Several final points should now be made. Firstly, many of the above steps concern themselves with diagnostics. One should liberally use certain routines like *specs*, *maplen* (*with sortl*), *showAssumption*, *showCommutator*, etc. so as to carefully monitor the general progression of the overall computation. Secondly, we stress that the generation/reduction processes are highly "non-linear" and hence final (and intermediate) system sizes and structures depend very much on intermediate steps and the order in which they are carried out. Again, we mention that the variable/operator orderings made at the beginning is important in the overall scheme.

4.3 Example 3 Purely magnetic spacetimes

There are no known vacuum spacetimes with a purely magnetic Weyl tensor and so it has been conjectured that no such spacetimes exist. Necessarily, the Weyl tensor must be of Petrov I, D, (excluding the trivial case of conformally flat). The conjecture has been established for Type D but despite extensive attempts by many researchers, the proof of the conjecture or the presentation of a counter example is yet to be achieved for Petrov I. The problem is deceptively difficult.

In this section, we will provide part of our proof of the conjecture, in the generic case of Type I. The complete proof with full details for all special sub-cases will be published elsewhere. Interestingly, this example centres on an unusual *STeM* environment consisting of a hybrid blend of null and orthonormal formalisms. We have found that if this problem is tackled purely in the null NP frame or in the orthonormal one, computational difficulties arise; expressions become large and unwieldy. This was easily seen with quick, short analyses using *STeM*. Part of the practical usefulness of *STeM* lies in its ability to allow the user to expeditiously "experiment" with various options composed of different formalisms and variables. Curiously, it is a hybrid choice, which we will describe below, that greatly facilitates the necessary computations. We note that for reasons of space limitations and aesthetic appeal we have chosen, via a Maple "*alias*" command to represent a complex conjugate by a preceding underscore instead of a trailing "_c". The user may wish to use other representations.

We begin with the NP environment where we initially choose the null tetrad to be the canonical one for a purely magnetic Weyl tensor of Petrov Type I, so that

$$\Psi_1 = \Psi_3 = 0, \Psi_0 = \Psi_4, \Psi_0 = -\bar{\Psi}_0, \Psi_2 = -\bar{\Psi}_2,$$

and since we have vacuum then

$$\Phi_{ab} = \Lambda = 0$$

We initialize the *STeM* environment as follows and indicate nonzero quantities.

```

>read 'c:/STeMV1R1.mpl':with(STeM):formalism(np):with(student):
>alias(_c1=c1_c,_c2=c2_c,_c3=c3_c,_c4=c4_c,_t=t_c,_k=k_c,_n=n_c,_p=p_c,
_a=a_c,_b=b_c,_g=g_c,_e=e_c,_l=l_c,_m=m_c,_s=s_c,_r=r_c):
>W1:=0:_W1:=0:W3:=0:_W3:=0:_W0:=-W0:_W4:=-W4:W4:=W0:_W2:=-W2:
>L:=0:R00:=0:R01:=0:R02:=0:R10:=0:R11:=0:R12:=0:R20:=0:R21:=0:R22:=0:
>stemassume({W0,W2}):

```

Next, we define an ordering on the spin coefficients based new variables $i1, \dots, i14$, $r1, \dots, r16$, and the curvature "compound" variables which will subsequently be introduced:

```
>stemdeforder([z1,z2,w6,w8,w9,w10,w13,w14,x9,x10,i6,i5,r6,r1,i1,r2,r3,i3,r4,i4,r5,r7,i7,r8,i8,r9,i9,r10,i10,r11,i11,r12,
i12,r13,r14,i13,i14,r15,r16]);
```

Based on extensive experience in using *STeM* over a period of more than 10 years, we have up to this point in time generally selected the "curvature type" variables to be in the highest ordered positions, which in this problem would amount to placing the w and x variables at the end of the ordering list. However, for this case it turns out that the lowest ordered positions is by far, the best. With these variables placed in the highest positions, one experiences significant issues regarding size and subsequent analysis of generated conditions. We stress that there are no hard-and-fast rules in deciding the order of the variables. This depends very much on the intricate structure of the system one is dealing with. As such we would recommend that the user should experiment a little with the ordering placed on variables before carrying out in-depth analysis on any problem. We now introduce the Ricci and Bianchi equations with complex conjugates, and combine.

```
>eins:=clean(map(eval,eqns(1,29))):ceins:=map(eval,map(conj,eins)):toteqs:=[op(eins),op(ceins)]:
```

The new operators $[E0, E1, E2, E3]$, are introduced, given an ordering, properties defined, and then replaced, by issuing the following commands:

```
>defineOperator('E0', E0=(D+V)):defineOperator('E1', E1=(D-V)):defineOperator('E2',E2=(X+Y)):
>defineOperator('E3', E3=(X-Y)):stemdefoper([E3, E2, E1, E0]):
>stemdefine(E0,type=[real]):stemdefine(E1,type=[real]):stemdefine(E2,type=[real]):stemdefine(E3,type=
[imag]):replaceOperator('D',D=(E0+E1)/2):replaceOperator('V', V=(E0-E1)/2):
>replaceOperator('X', X=(E2+E3)/2):replaceOperator('Y', Y=(E2-E3)/2):
>
```

We note that because of space limitations, we have suppressed the obvious Maple output. Curiously, we have defined the new operators with a "lopsided" factor of 2. Presumably, for mainly aesthetic reasons one would want to see the " $\sqrt{2}$ " introduced instead so as to mimic the correspondence with orthonormal operators as closely as possible. However, if one does this then Maple is forced to work with this irrational quantity and as a consequence, one finds that the performance is reduced.

Before proceeding further with the investigation of this problem, we would like to introduce new notions of *simple* and *compound* rotations. Now the spin coefficients $\{\kappa, \sigma, \rho, \pi, \tau, \mu, \lambda, \nu\}$ are proper weighted quantities but $\{\alpha, \beta, \gamma, \epsilon\}$ are not since under the spin-boost null rotations, $\{\tilde{l}^\mu = A^{-1}l^\mu, \tilde{n}^\mu = An^\mu, \tilde{m}^\mu = e^{i\theta}m^\mu\}$, these quantities in general contain both derivatives of A and θ in their transformations. We say that these spin coefficients involve *compound* rotations. However consider the combinations $\{\alpha + \bar{\beta}, \alpha - \bar{\beta}, \epsilon + \bar{\epsilon}, \epsilon - \bar{\epsilon}, \gamma + \bar{\gamma}, \gamma - \bar{\gamma}\}$. They transform as

$$\begin{aligned}\tilde{\alpha} + \tilde{\bar{\beta}} &= e^{-i\theta} \{\alpha + \bar{\beta} - A^{-1}\bar{\delta}A\}, \tilde{\alpha} - \tilde{\bar{\beta}} = e^{-i\theta} \{\alpha - \bar{\beta} + i\bar{\delta}\theta\}, \\ \tilde{\epsilon} + \tilde{\bar{\epsilon}} &= A^{-1}(\epsilon + \bar{\epsilon}) - A^{-2}DA, \tilde{\epsilon} - \tilde{\bar{\epsilon}} = A^{-1}(\epsilon - \bar{\epsilon}) + iA^{-1}D\theta, \\ \tilde{\gamma} + \tilde{\bar{\gamma}} &= A(\gamma + \bar{\gamma}) - A^{-2}DA, \tilde{\gamma} - \tilde{\bar{\gamma}} = A(\gamma - \bar{\gamma}) - iA\Delta\theta.\end{aligned}$$

Since the transformations of these quantities involve derivatives of A or θ but **not simultaneously** both, we say that they involve *simple* boost/rotations or are just *simple*. Therefore, we would like to focus our attention on *simple* real and purely imaginary combinations. Hence we arrive at the following variable transformations:

$$\begin{aligned}r_1 &= \alpha + \bar{\beta} + \bar{\alpha} + \beta, i_1 = \bar{\alpha} + \beta - \alpha - \bar{\beta}, \\ r_2 &= \bar{\alpha} - \beta + \alpha - \bar{\beta}, i_2 = \bar{\alpha} - \beta - \alpha + \bar{\beta}, \\ r_3 &= \epsilon + \bar{\epsilon} + \gamma + \bar{\gamma}, i_3 = \epsilon - \bar{\epsilon} + \bar{\gamma} - \gamma, \\ r_4 &= \gamma + \bar{\gamma} - \epsilon - \bar{\epsilon}, i_4 = -\gamma + \bar{\gamma} - \epsilon + \bar{\epsilon}.\end{aligned}$$

Next, we focus on the remaining spin coefficients. Now $\{\rho, \mu\}$ involve only the boost, and $\{\sigma, \bar{\lambda}\}$ both have the same spin factor. So continuing with the same general idea, by not mixing the different types, at this stage yet, and taking real and purely imaginary parts, modulo 2, we arrive at the following new variables for $\rho, \mu, \sigma - \bar{\lambda}, \sigma + \bar{\lambda}$:

$$\begin{aligned}r_5 &= \rho + \bar{\mu} + \bar{\rho} + \mu, i_5 = \bar{\rho} + \mu - \rho - \bar{\mu}, \\ r_6 &= \rho - \bar{\mu} + \bar{\rho} - \mu, i_6 = \bar{\rho} - \mu - \rho + \bar{\mu}, \\ r_7 &= \sigma + \bar{\lambda} + \bar{\sigma} + \lambda, i_7 = \bar{\sigma} + \lambda - \sigma - \bar{\lambda}, \\ r_8 &= \sigma - \bar{\lambda} + \bar{\sigma} - \lambda, i_8 = \bar{\sigma} - \lambda - \sigma + \bar{\lambda}.\end{aligned}$$

This leaves the remaining spin coefficients $\{\kappa, \tau, \pi, \nu\}$. By looking at similar spin transformation properties, we should consider the basic combinations: $\{\tau + \bar{\pi}, \tau - \bar{\pi}, \kappa + \bar{\nu}, \kappa - \bar{\nu}\}$. Thus we take these complex combinations, and form their real and purely imaginary parts, modulo 2,

$$\begin{aligned} c_1 &= \kappa + \bar{\nu} + \tau + \bar{\pi}, c_2 = \tau + \bar{\pi} - \kappa - \bar{\nu}, \\ c_3 &= \bar{\nu} - \kappa + \bar{\pi} - \tau, c_4 = \bar{\pi} - \tau + \kappa - \bar{\nu}, \\ r_9 &= c_1 + \bar{c}_1, i_9 = c_1 - \bar{c}_1, \\ r_{10} &= c_2 + \bar{c}_2, i_{10} = c_2 - \bar{c}_2, \\ r_{11} &= c_3 + \bar{c}_3, i_{11} = c_3 - \bar{c}_3, \\ r_{12} &= c_4 + \bar{c}_4, i_{12} = c_4 - \bar{c}_4, \end{aligned}$$

Further, we find that it is advantageous to make the following additional changes:

$$\begin{aligned} i_5 &= (i_{13} - i_{14})/2, i_7 = (i_{13} + i_{14})/2, r_4 = (r_{14} - r_{13} + 2r_6)/4, \\ r_7 &= (r_{15} + r_{16})/2, r_8 = (r_{13} + r_{14})/2, r_5 = (r_{15} - r_{16})/2. \end{aligned}$$

Interestingly, these new variables correspond to the orthonormal combinations, modulo a numerical factor, $n_{11} + n_{22} - n_{33}$, $n_{11} - n_{22} + n_{33}$, $\theta_{11} - \theta_{22}$, $\theta_{11} - \theta_{33}$, $a_1 + n_{23}$, $a_1 - n_{23}$, similarly for the others. These combinations are significantly simpler than the corresponding NP ones. As yet, it is not clear to us how these and the other variable combinations can be found while completely within the orthonormal formalism.

Our input is, with the addition of z_1 and z_2 , as follows.

```
>set1:={z1=W0+3*W2,z2=W0-3*W2}
>set2:={r1 = _a + b + a + _b, i1 = _a + b - a - _b, r2 = _a - b + a - _b, i2 = _a - b - a + _b}:
>set3:={r3 = _g+e+_e+g, i3 = _g+e-g-_e, r4 = _g-e+g-_e, i4 = _g-e-g+_e}:
>set4:={r5=_r + m + r + _m, i5=_r + m - r - _m, r6=_r - m + r - _m, i6=_r - m - r + _m}:
>set5:={ c1 = k+t+_p+_n, c2 = t-k+_p-_n, c3 = _n-k+_p-t, c4 = _p-t-_n+k}:
>set5a=map(conj,set5):
>set6:={r7 = _s+l+s+_l, i7 = _s+l-s-_l, r8 = _s-l+s-_l, i8 = _s-l-s+_l}:
>set7:={c1=(r9+i9)/2, _c1=(r9-i9)/2, c2=(r10+i10)/2, _c2=(r10-i10)/2, c3=(r11+i11)/2, _c3=(r11-i11)/2,
c4=(r12+i12)/2, _c4=(r12-i12)/2, i5 = (1/2)*i13-(1/2)*i14, i7 = (1/2)*i14+(1/2)*i13,
r4 = (1/4)*r14-(1/4)*r13+(1/2)*r6, r8 = (1/2)*r13+(1/2)*r14, r5 = (1/2)*r15-(1/2)*r16, r7 = (1/2)*r16+(1/2)*r15}:
```

Our semi-final variable set is $\{r1, r2, r3, r6, r9, \dots, r16, i1, \dots, i4, i6, i8, \dots, i14, z1, z2\}$, correctly yielding 24 independent real/imaginary functions. We introduce the properties of the new variables, into the session by two small "do" loops with just one entry:

```
>for j from 1 to 16 do stemdefine(r||j,type=['real']):od:
>for j from 1 to 14 do stemdefine(i||j,type=['imag']):od:
```

and

```
>stemdefine(z1,type=['imag']):stemdefine(z2,type=['imag']):
```

We need one final change which scales the variables $\{i6, i8, i9, i10, i13, i14, r9, r10\}$ by the curvature components and by $\{i1, i3, i4, r1\}$.

```
>set8:={i10 = w10*i1/z2, i13 = i3*w13/z1, i14 = i3*w14/z2, i6 = w6*i4/(z2*z1), i8 = w8*i4/(z2*z1),
i9 = w9*i1/z1, r10 = x10*r1/z1, r9 = r1*x9/z2}:
>stemdefine(z1,type=['imag']):stemdefine(z2,type=['imag']):stemdefine(x9,type=['imag']):stemdefine(x10,type=['imag']):
>stemdefine(w13,type=['imag']):stemdefine(w14,type=['imag']):stemdefine(w6,type=['real']):stemdefine(w8,type=['real']):
>stemdefine(w9,type=['imag']):stemdefine(w10,type=['imag']):
```

We can easily compare our new variables in terms of the standard orthonormal ones as follows, for example:

```
>factor(solve(map(convertNP2Orth,set2),{r1,i1,r2,i2}));
```

$$\{i1 = I\sqrt{2}(s13-w2), i2 = I\sqrt{2}(a3-n12), r1 = \sqrt{2}(s12+w3), r2 = (a2+n13)\sqrt{2}\}$$

```
>factor(solve(map(convertNP2Orth,set6),{r7,i7,r8,i8}));
```

$$\{i7 = I\sqrt{2}(n22-n33), i8 = (2I)\sqrt{2}s23, r7 = 2\sqrt{2}n23, r8 = -(H2-H3)\sqrt{2}\}$$

A check of all the new variables shows that they are intriguingly simple combinations of the orthonormal ones. However, curiously, we have found that, for this problem, a direct application of the standard orthonormal system leads to insurmountable difficulties.

We complete the setup of the *STeM* environment by making a global total switch from the NP standard variables to our new ones.

```
>solve({op(set1),op(set2),op(set3),op(set4),op(set5),op(set5a),op(set6)},
{a,_a,b,_b,g,_g,p,_p,t,_t,e,_e,n,_n,m,_m,k,_k,l,_l,r,_r,s,_s,W0,W2}):assign(%):
>assign(set7):assign(set8):
```

The stage is now set for the analysis of the complete set of the extended field equations, with the application of *stemnormal*. We display some of the output with decision choices.

```
>toteqsa:=stemnormal(toteqs,switch=[algebraic,factor]):
```

"Obtain algebraic equation"

```
i1*w10-i1*w9+r1*x10-r1*x9
```

Substitute this algebraic equation into the system? (y,n) y

"Obtain algebraic equation"

```
r1*(x10-x9)
```

Substitute this algebraic equation into the system? (y,n) y

factor , 1, : , r1

factor , 2, : , x10 - x9

Use factor number?(ter=terminate) 2

"Obtain algebraic equation"

```
i1
```

Substitute this algebraic equation into the system? (y,n) y

"Obtain algebraic equation"

```
x10-2*z1-2*z2
```

Substitute this algebraic equation into the system? (y,n) y

"Obtain algebraic equation"

```
x9-2*z1-2*z2
```

Substitute this algebraic equation into the system? (y,n) y

"Obtain algebraic equation"

```
i3*w13*z1*z2+i3*w14*z1*z2+i4*w6*z1-i4*w6*z2-i4*w8*z1-i4*w8*z2
```

Substitute this algebraic equation into the system? (y,n) y

"Obtain algebraic equation"

```
i3*(w13+w14)
```

Substitute this algebraic equation into the system? (y,n) y

Therefore there are 24 polynomial conditions of the indicated lengths. Even though the system *totpols* is quite simple we can in fact simplify it much further, using the routines *minimisePoly* and *reduceMinSys*, as follows. Before that, we should query *STeM* to indicate what assumptions have been made.

```
>showAssumption();
```

```
Non-zero factors
i3 ≠ 0
r1 ≠ 0
Non-zero denominators
r1 ≠ 0
w6 ≠ 0
w8 ≠ 0
z1 ≠ 0
z2 ≠ 0
w6 - w8 ≠ 0
-z1 - z2 ≠ 0
-w10 + w9 ≠ 0
-w6 z1 + w6 z2 + w8 z1 + w8 z2 ≠ 0
```

```
>totpolb:=minimisePoly(totpols):
>totpolc:=reduceMinSys(totpolb):
>findops(totpolc);
```

11, 12

```
>totpolc[11]:=totpolc[11]/z1/z2;
```

```
totpolc[11] := 3*i11*r1*z2-3*i12*r1*z1-6*i2*r1*z1+6*i2*r1*z2-6*i3*r14*z1+2*z1^2-8*z1*z2
```

```
>totpolc[12]:=totpolc[12]/(x9-2*z1);
```

```
totpolc[12] := i11*r1*x9+2*i11*r1*z1-2*i12*r1*z1+2*i2*r1*x9-2*i3*r13*z1-i3*r14*x9-2*i3*r14*z1
```

```
>showStatistic(totpolc);
```

```
Number of equations is , 13
Number of polynomial equations is , 13
Number of differential equations is , 0
Equations' length are
[2, 2, 12, 12, 12, 16, 92, 104, 104, 115, 136, 140, 167]
Equations' LHS are
[none, none, none, none, none, none, none, none, none, none,
none, none, none]
```

We see that the system of polynomials has been reduced considerably. Finally, we use the routine *PolynomialSystem*, in Maple, to solve *totpolc*.

```
> with(SolveTools):
>sola:=PolynomialSystem(totpolc,{z1<>0,z2<>0});
```

```
sola := {i1 = 0, i11 = i11, i12 = i12, i2 = i2, i3 = i3, i4 = 0, r1 = 2z1/(2i2+i12), r13 = -2z1/i3,
r14 = -2z1(-i12+i11)/(i3(2i2+i12)), w13 = 2z1, w14 = -2z1, x10 = -2z1, x9 = -2z1, z1 = z1, z2 = -2z1}
```

Note that $2i_2 + i_{12} \neq 0$ since $z_1 z_2 \neq 0$. There is a solution to the conditions obtained so we are not quite done yet. To finish this example, we simply apply *E0* to $z_2 = -2z_1$.

```
>fsubs(sola,stemreduce(E0(z2=-2*z1),toteqsa));
```

$-12z_1^2/i_3$

Thus the proof for this generic case is complete. We will stop here. There are of course, a number of special cases, that need to be analyzed, as given by $r_1 i_3 w_6 w_8 (w_6 - w_8)(w_9 - w_{10})(-w_6 z_1 + w_6 z_2 + w_8 z_1 + w_8 z_2) = 0$. It turns out that some of these subcases are more involved. However, the overall resulting systems, that we do find, have the desirable feature that there occurs a good measure of variable-separation in the constituent conditions which assists the analysis considerably, though, admittedly, there results a large number of branching cases.

It is apparent from the above work, that it was not necessary, at any stage, to explicitly display the complete system. There are a number of routines in *STeM* which assist the user to extract essentially a great deal of the important/relevant information, of any given system without displaying it.

As a final point, it is conceivable that the new variables/system introduced here may prove to be useful for other applications. This matter is currently under investigation.

5 Comments

In general relativity one usually encounters large complex systems of equations. Hence, to assist the user in the investigation of such systems, *STeM* provides interactive tools to assist in the computations and to help make important decisions at certain critical points about how to proceed with the analysis of particular systems. We emphasize that frequently, the best approach, which fundamentally relies on the ordering of the variables that affect *STeM*'s reduction routines, is not immediately obvious and therefore some experimentation may be required.

In assembling the different parts of *STeM*, we have been required to input hundreds of equations from various sources. Clearly, the challenge has been to provide consistent, in the sense of signature, duals and object definitions, systems of equations without typographical error. We have done our best to try and avoid errors by performing very extensive cross-checking but however being that this is the first release of *STeM*, we recommend that all users exercise appropriate caution regarding the accuracy of any equations. Separate independent checks is advised. We hope that through the widespread use of this package, $\pm$ any errors that may be present will be found. In such cases, we would much appreciate receiving notification of these issues so that they may be addressed. In the second part exposition of *STeM*, we will demonstrate by examining further applications, the functionality of the remaining routines that we have not covered here. As a final point, in the near future, if there is sufficient interest, we will develop a version of *STeM* for Mathematica.

6 Acknowledgments

First of all, we would like to thank N.Van den Bergh for providing us with most of the orthonormal equations that are in *STeM*. It is his notation and conventions, except for the tetrad rotation coefficients renaming, that we have implemented in the orthonormal part of our package. We would also thank him for providing the Maple worksheets containing the expressions of the defining basic variables and basic integrability conditions/EFE, which are included in *STeM*, and for his continued encouragement and suggestions throughout various aspects of this work. For the NP and GHP Killing equations, with resulting integrability conditions, (these were independently verified by *STeM*) we thank J.D. Steele.

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